

Brieuc FRÉNAIS

Curriculum Vitae

Mines Nancy (ENSMN)
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Education

- Sep. 2021 – **PhD in Mathematics**, IRMA (Strasbourg, France).
- Dec. 2024 *Stochastic branching processes with selection*, under supervision of Jean Bérard and Lionel Lenôtre, defended on December 20, 2024 (HAL).
- 2020–2021 **Master of Mathematics (Probability theory)**, Université de Strasbourg.
- 2017–2021 **Magistère de Mathématiques**, École Normale Supérieure de Rennes, Université de Rennes I.
Bachelor's degree in 2018, Master's degree in 2020. Agrégation de Mathématiques in 2020.
- 2015–2017 **CPGE MPSI-MP***, Lycée Clemenceau (Nantes, France).
- 2015 **Baccalauréat Scientifique**, Lycée Clemenceau (Nantes, France).

Research & Teaching Experience

Ecole des Mines, Nancy (ENSMN) & IECL, Université de Lorraine

- Sep. 2025 – **PRAG (Teaching professor)**.
present Full teaching position in engineer school. Responsible for the Mathematical Engineering option. Involved in student recruitment and academic advising.
- Sep. 2024 – **ATER (non-tenured teaching and research position)**.
Aug. 2025 Continuation of my PhD, as well as teaching to engineer-students on various topics.
- IRMA, Université de Strasbourg
- Sep. 2021 – **PhD in Mathematics**, *Stochastic branching processes with selection*.
Dec. 2024 Under supervision of Jean Bérard and Lionel Lenôtre, defended on December 20, 2024.
- Sep. 2021 – **Teaching assignment**.
June 2024 Teaching classes from Bachelor's to Master's degree students (mainly on Probability).
- Sep. 2022 – **Co-organizer of PhD students' seminar**.
Sep. 2023
- Summer 2022 **Summer School CEMRACS 2022**, CIRM (Marseille, France).
One week summer school, five weeks working on a research project about a mean field control problem.
- Oral interrogations in CPGE
- 2020–2021 **Mathematics colles in MPSI & PCSI**, Lycée Kléber (Strasbourg, France).
- 2018 **Mathematics colles in MPSI**, Lycée Clemenceau (Nantes, France).
- Other experiences
- April 2021 – **Research internship**, IRMA (Strasbourg, France).
- August 2021 Working on the hydrodynamic limit of the N-branching Brownian motion, under supervision of Jean Bérard and Lionel Lenôtre.
- Summer 2019 **Research internship**, Laboratoire Jean Leray (Nantes, France).
Working on the connective constant of the hexagonal lattice, under supervision of Nicolas Pétrélis.
- Summer 2018 **Research internship**, Laboratoire Jean Leray (Nantes, France).
Working on resonance and periodicity phenomena for some dynamic systems, under supervision of Benoît Grébert.
- 2015–2017 **Private teacher for highschool students**.

Computer & linguistic skills

Programming LaTeX, Python, Keras, Tensorflow, R, HTML/CSS
Languages French (mother tongue), English (written, read, spoken), German (basic notions)

Publications

Conference proceedings

Nov. 2024 **Deep Learning for Mean Field Optimal Transport**, S. Baudelet, B. Frénais, M. Laurière, A. Machtalay and Y. Zhu, In **ESAIM: ProcS, Vol. 77** (HAL)(DOI).

Submitted papers & pre-prints

Nov. 2023 **The comonotone flow of a stochastically monotone Feller process on the real line**, J. Bérard and B. Frénais, (HAL).

Nov. 2023 **Hydrodynamic limit of N-branching Markov processes**, J. Bérard and B. Frénais, (HAL).

Talks

Nov. 2025 **Hydrodynamic limit of a branching Markov process with selection**, IRN ReaDiNet's Workshop (Obernai).

June 2025 **Limite hydrodynamique de processus de branchement avec sélection**, Probability&Statistics seminar (IECL, Nancy).

May 2025 **Limites hydrodynamiques, problèmes à frontière libre et temps de passage**, PhD students' seminar (IECL, Nancy).

July 2024 **Problème inverse du premier temps de passage et problème à frontière libre**, MATHeOR Days 2024 (Sainte-Marie-aux-Mines).

June 2024 **Limite hydrodynamique d'un processus de branchement avec sélection**, Journées de Probabilités 2024 (Bordeaux).

June 2024 **Limite hydrodynamique d'un processus de branchement avec sélection**, Probability seminar (IRMAR, Rennes).

March 2024 **Limite hydrodynamique de processus de Markov branchants avec sélection**, Mathematics seminar (IRIMAS, Mulhouse).

Nov. 2023 **Limite hydrodynamique de processus de Markov branchants avec sélection**, Stochastic calculus seminar (IRMA, Strasbourg).

Nov. 2023 **Limite hydrodynamique d'un processus de branchement-sélection**, PhD students' seminar (IRMA, Strasbourg).

October 2023 **Méthodes de machine learning pour le problème de transport optimal en champ moyen**, Workshop on Optimal Transport (IRIMAS, Mulhouse).

Feb. 2023 **Méthodes de machine learning pour le problème de transport optimal en champ moyen**, PhD students' seminar (IRMA, Strasbourg).

Aug. 2022 **Numerical methods for mean field optimal transport**, CEMRACS project presentation (Marseille).

Nov. 2021 **Vitesse de croissance exponentielle des chemins auto-évitant sur le réseau hexagonal**, PhD students' seminar (IRMA, Strasbourg).

Scientific mediation

April 2023 **Rendez-vous des Jeunes Mathématiciennes et Informatiennes**, IRMA, Université de Strasbourg, Supervision of a mathematical activity.

April 2023 **Tournoi Français des Jeunes Mathématiciennes et Mathématiciens**, UFR de mathématique et d'informatique, Strasbourg, Jury for the regional tournament.

Feb. 2022 **Rendez-vous des Jeunes Mathématiciennes et Informaticiennes**, *IRMA, Université de Strasbourg*, Supervision of a research workshop and a mathematical activity.